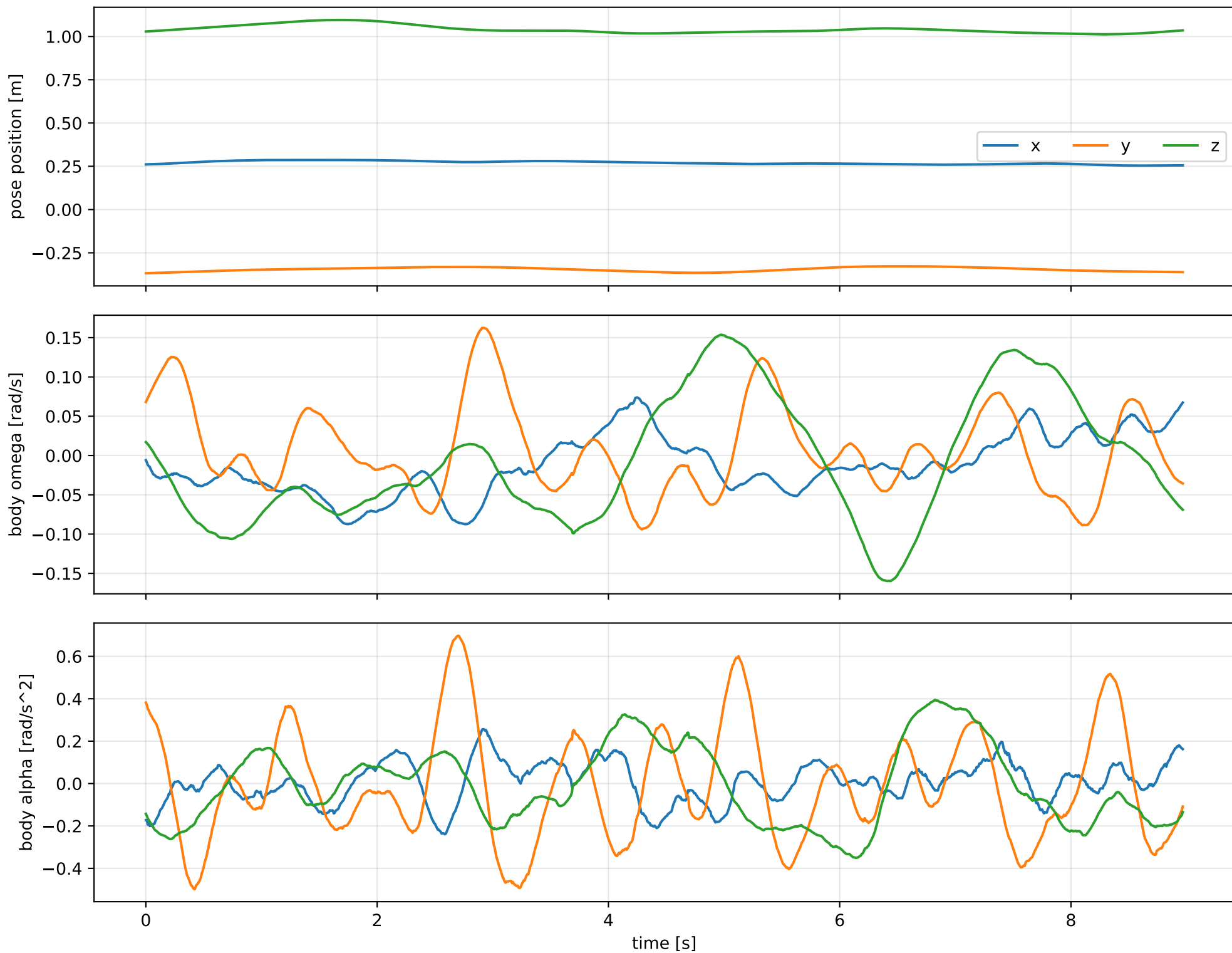
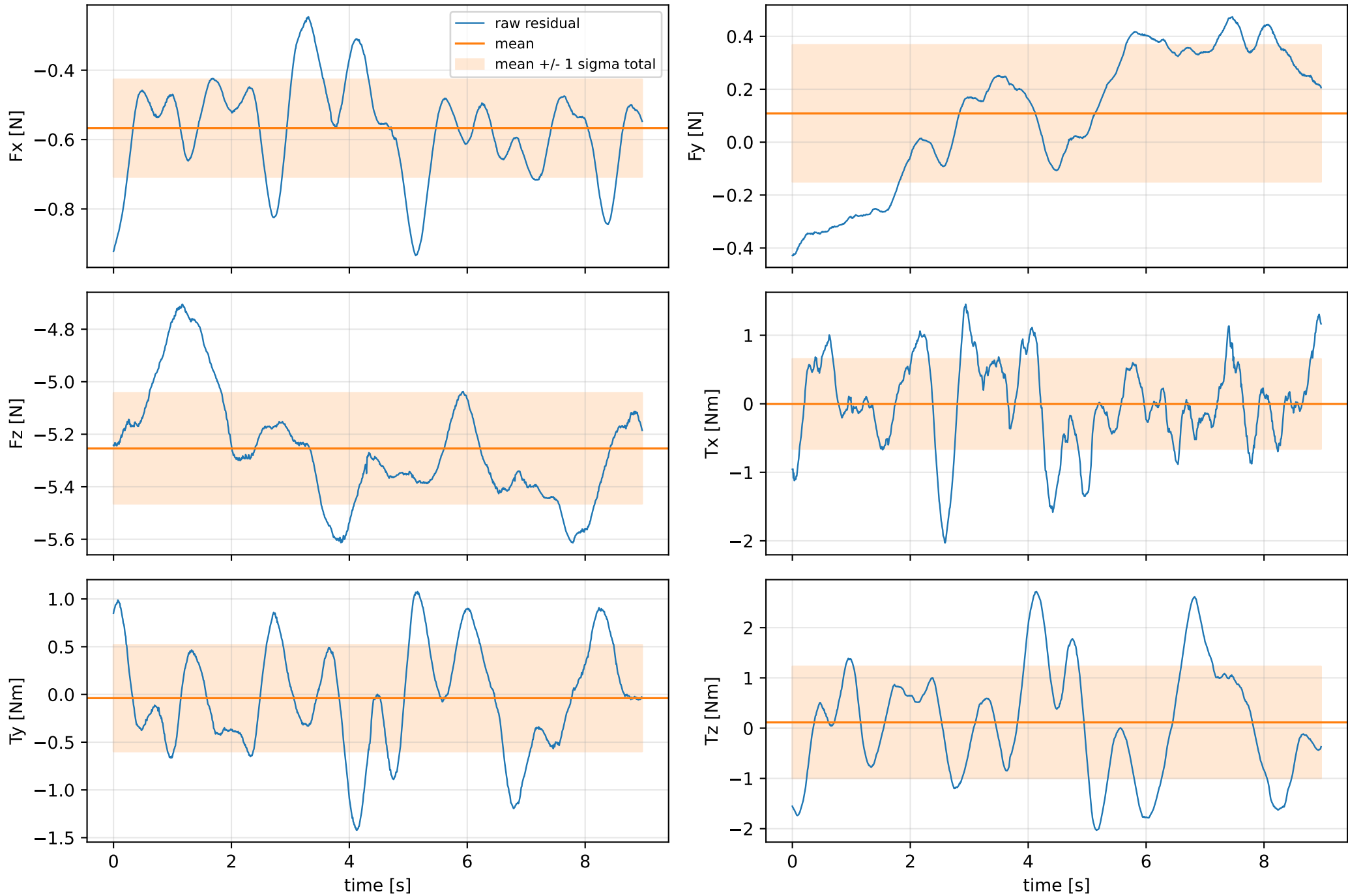


force_over_mass_all_nominal: SG trajectory and derived motion



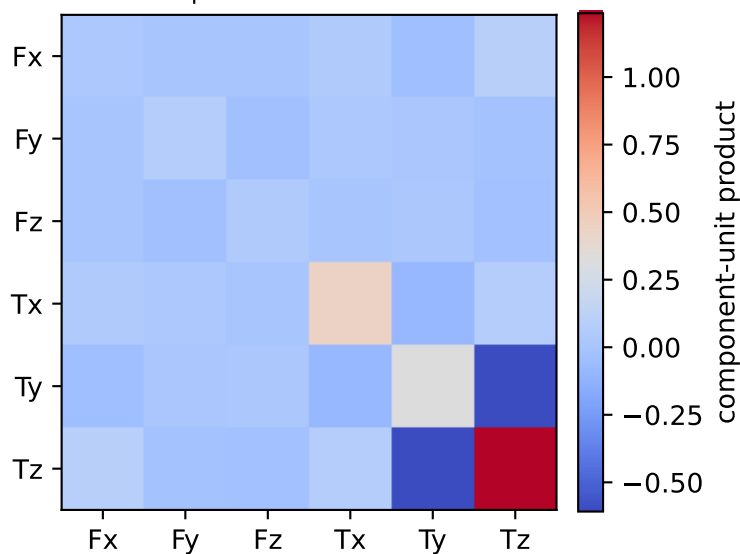
force_over_mass_all_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



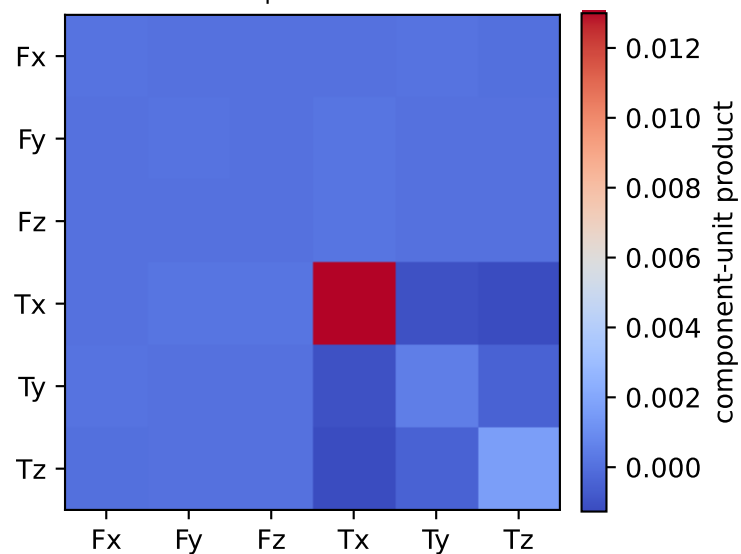
vector RMS about zero: force 5.2978 N, torque 1.4111 Nm; component std about mean: force [0.1405 0.2585 0.2109], torque [0.6564 0.5581 1.1118]

force_over_mass_all_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.003 0.011 0.017 0.091 0.42 1.551]

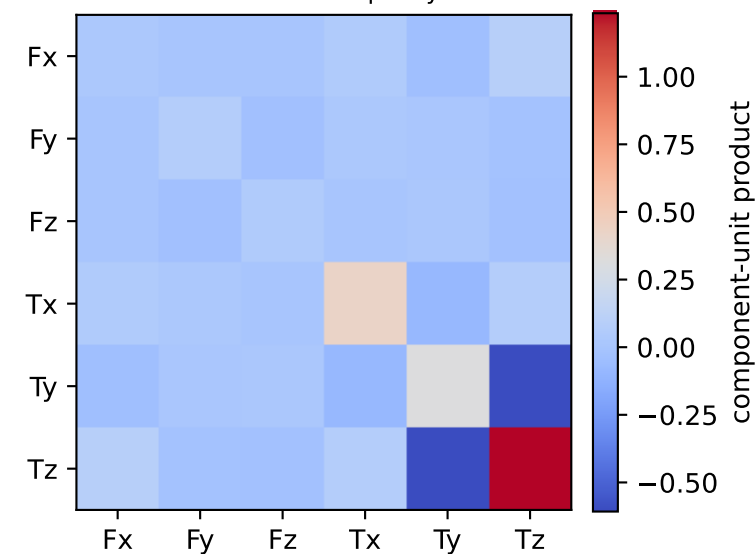
empirical total covariance



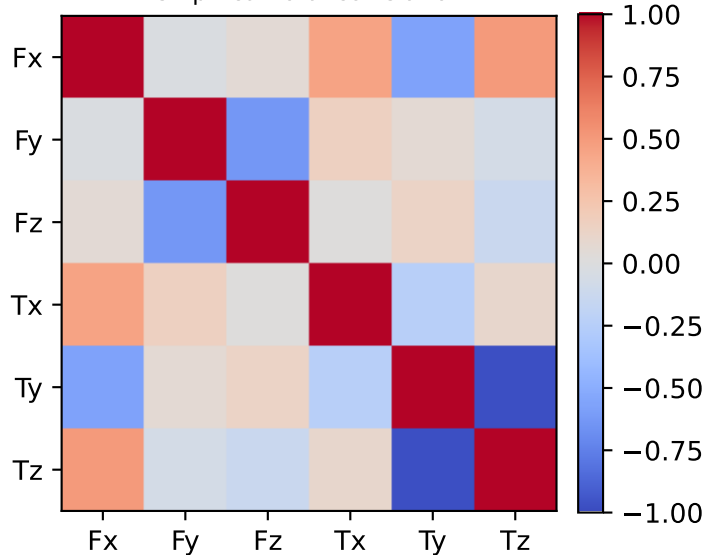
mean full-SG prediction covariance



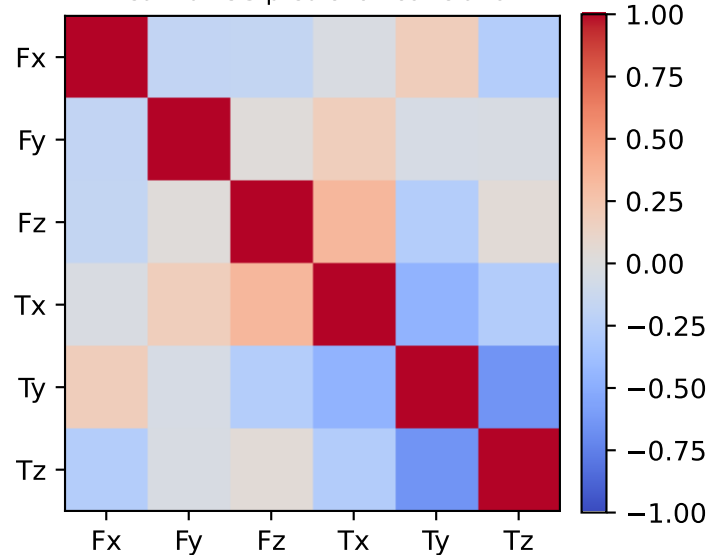
PSD excess model discrepancy covariance



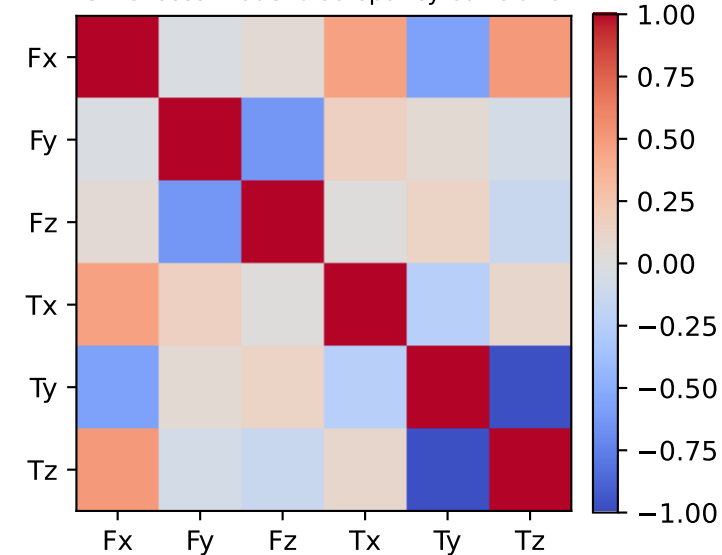
empirical total correlation



mean full-SG prediction correlation

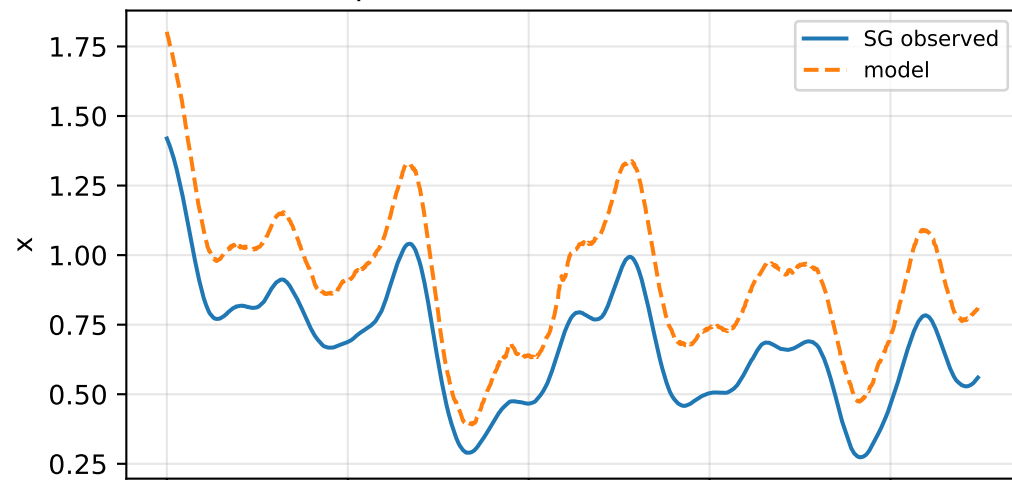


PSD excess model discrepancy correlation

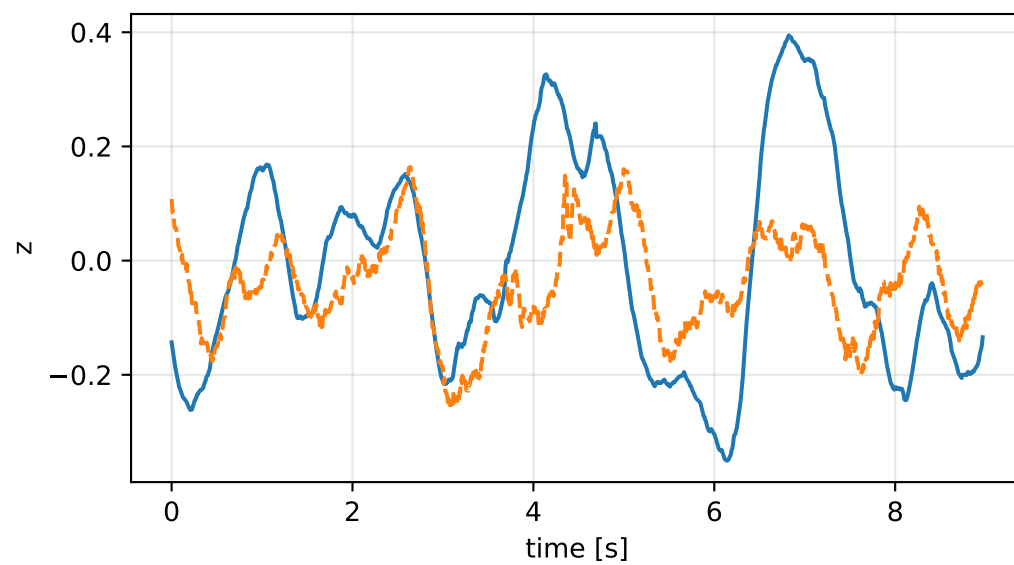
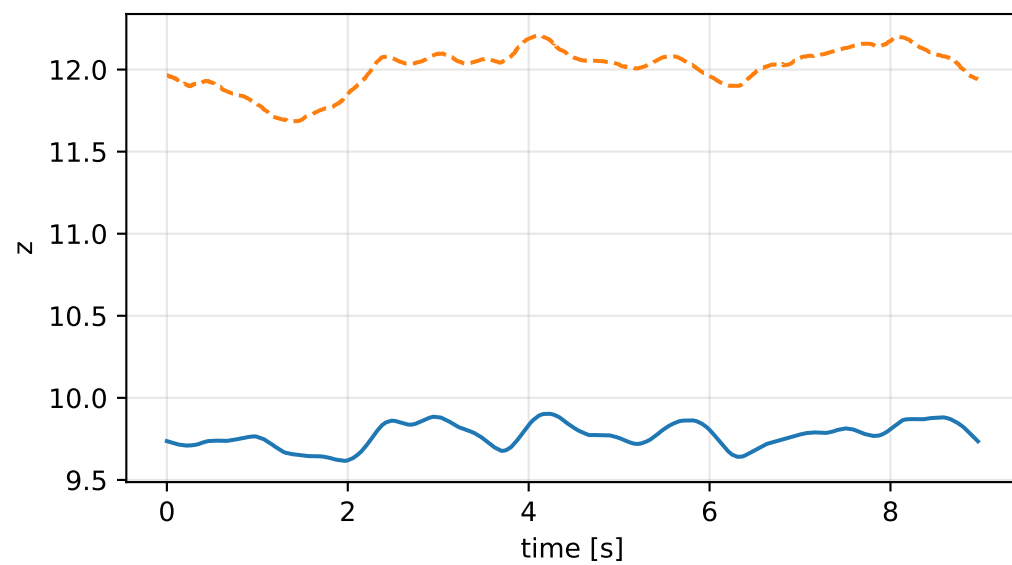
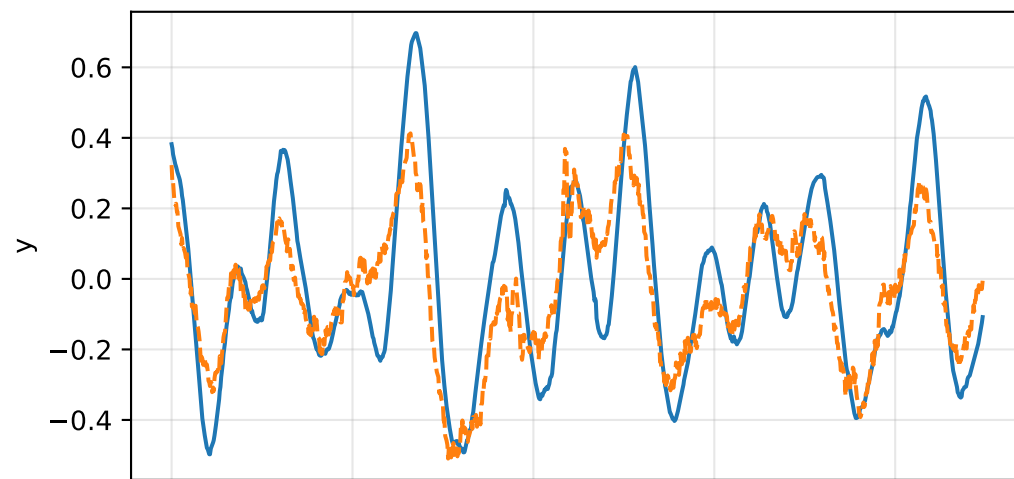
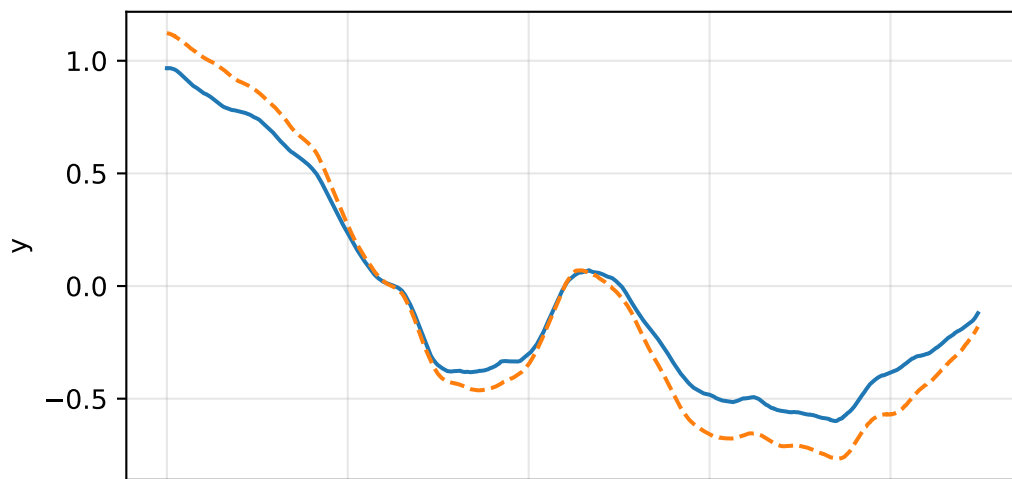
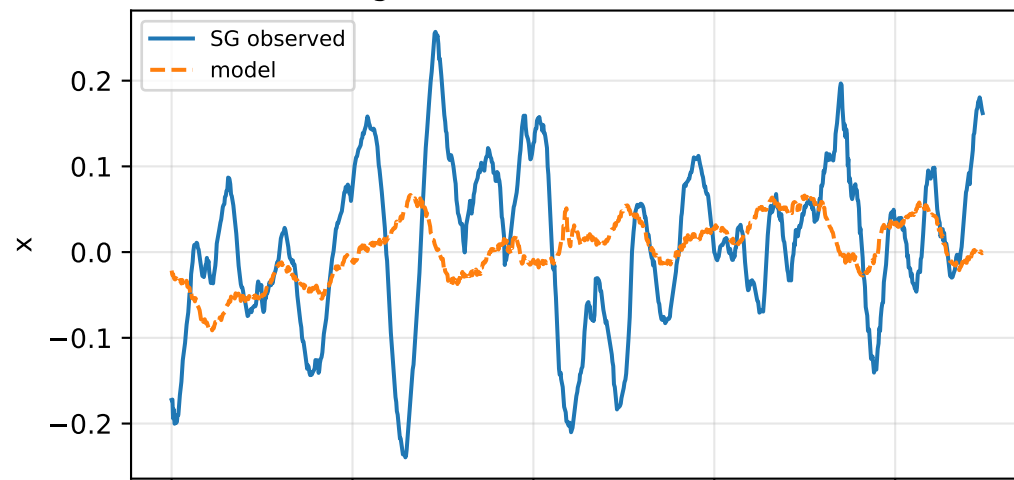


force_over_mass_all_nominal: acceleration residual objective

specific acceleration [m/s^2]

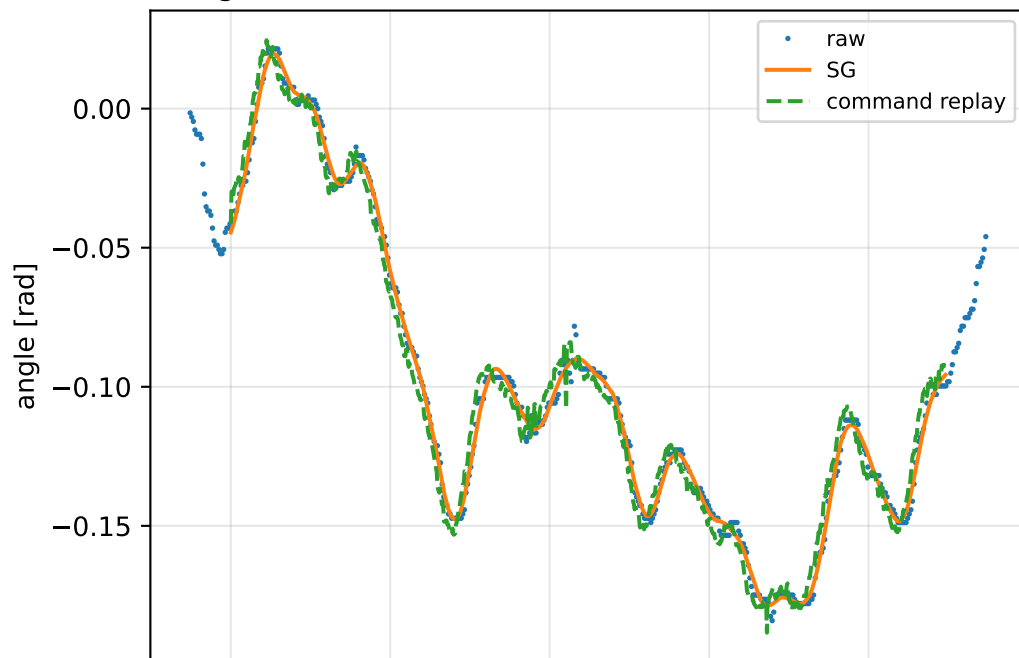


angular acceleration [rad/s^2]

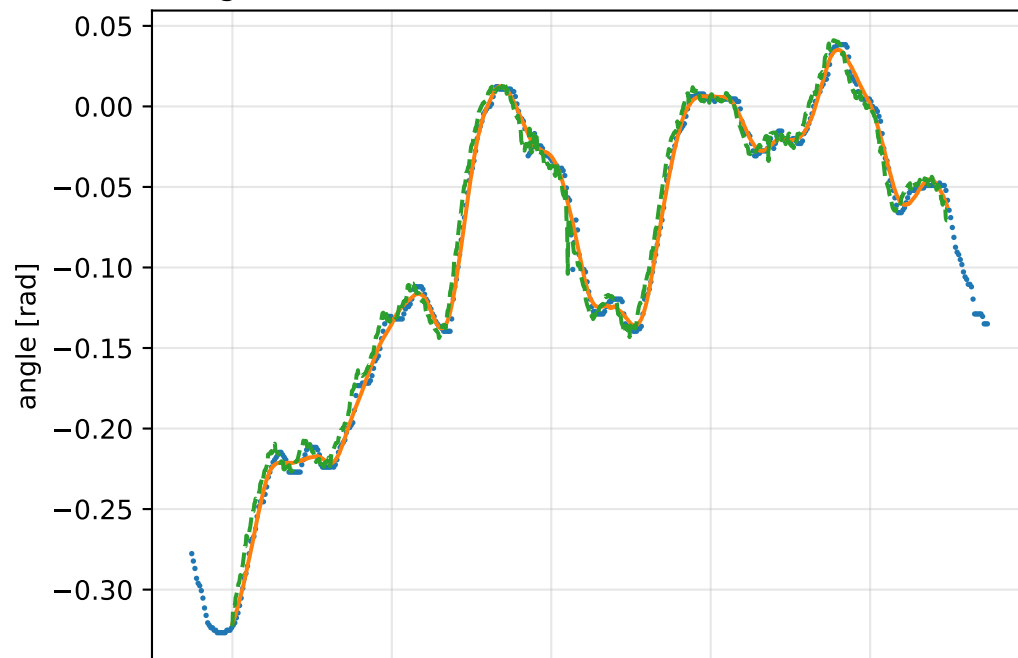


force_over_mass_all_nominal: gimbal measurement audit (objective=measured_sg)

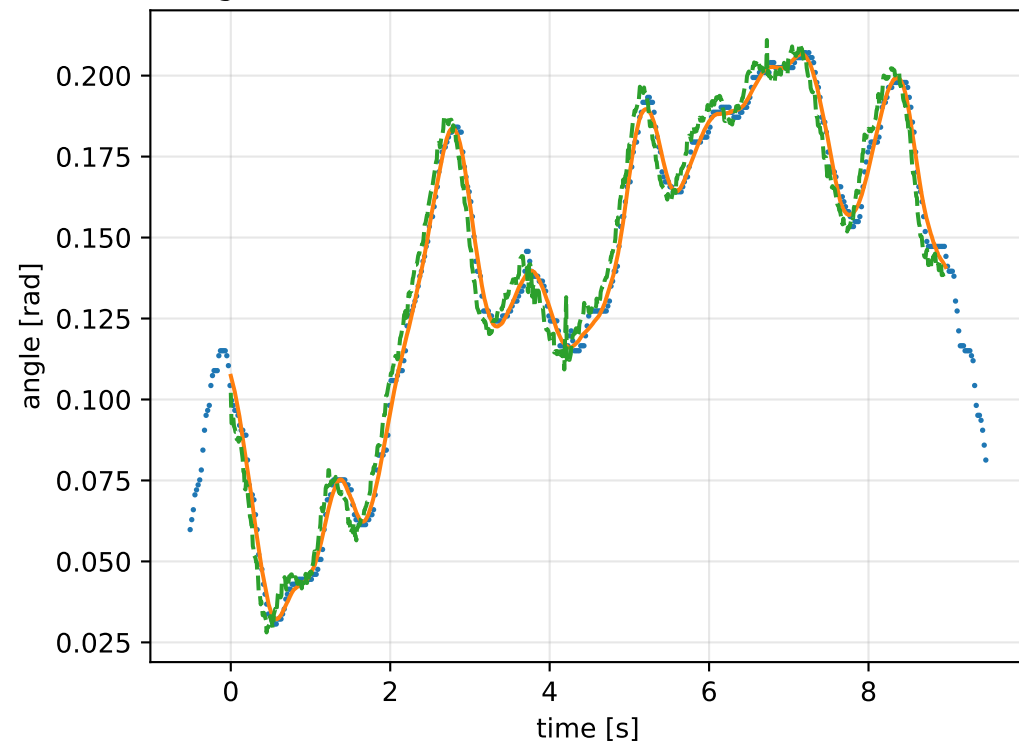
gimbal 1: RMSE=0.006345, max=0.01494 rad



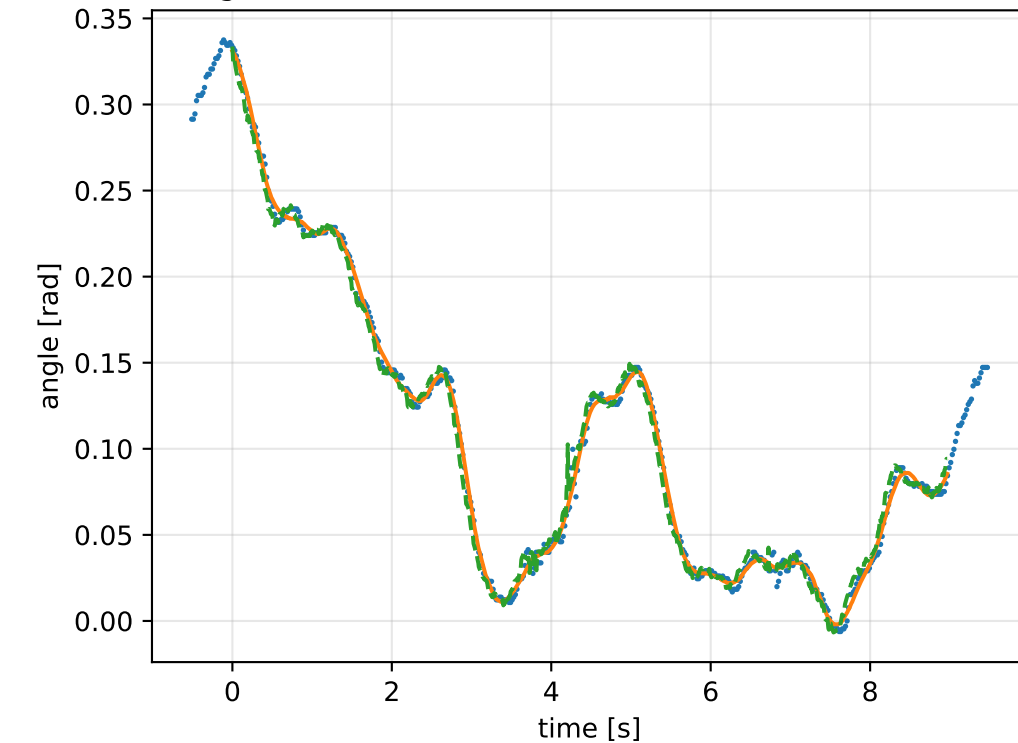
gimbal 2: RMSE=0.008835, max=0.04462 rad



gimbal 3: RMSE=0.007019, max=0.01611 rad

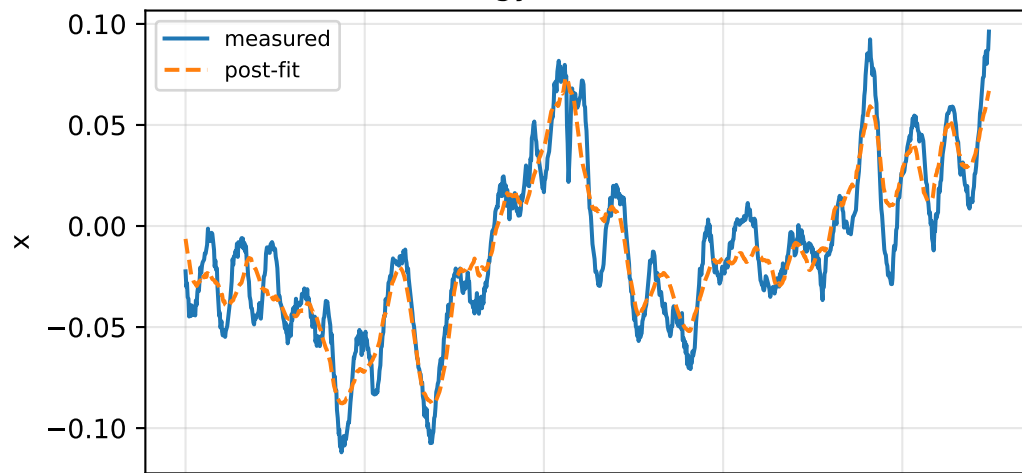


gimbal 4: RMSE=0.006321, max=0.03642 rad

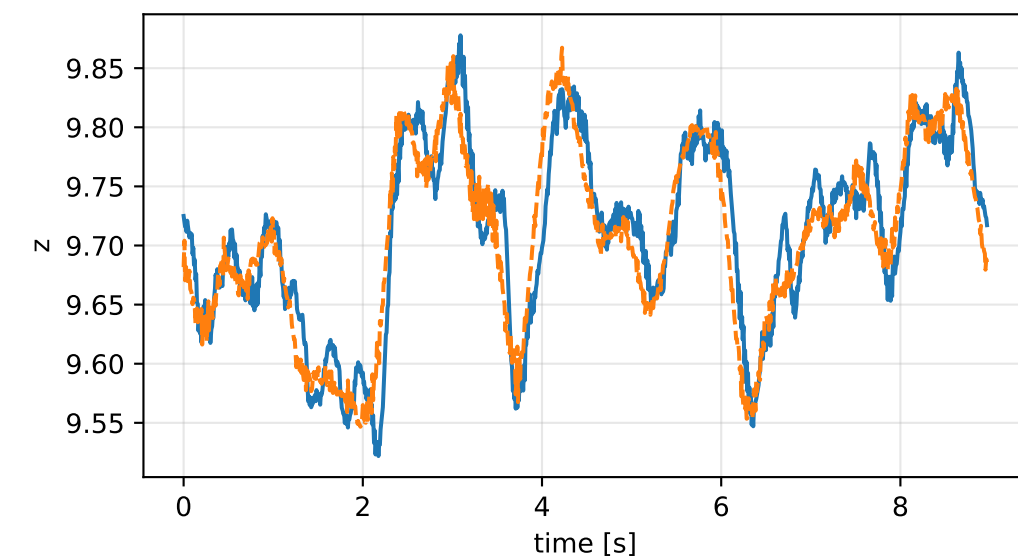
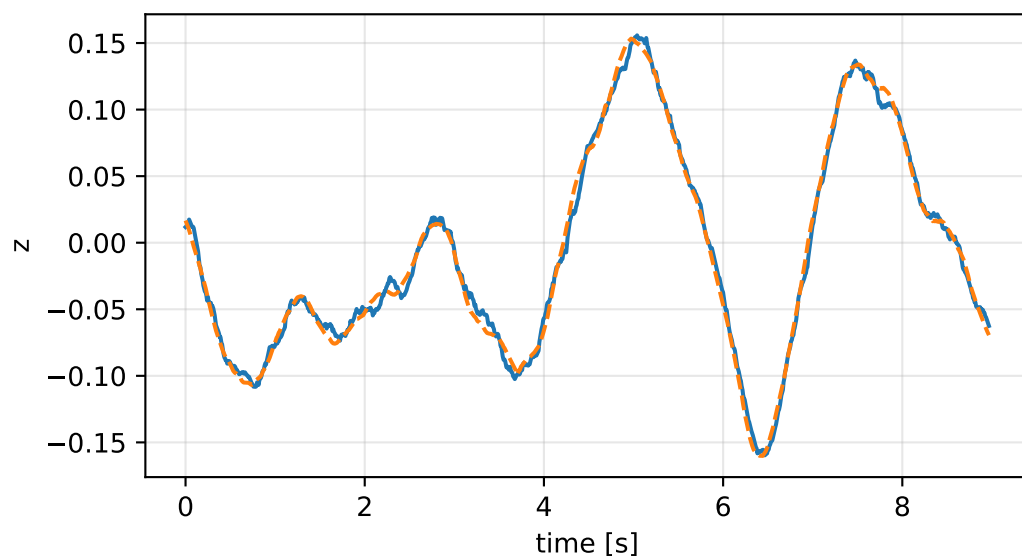
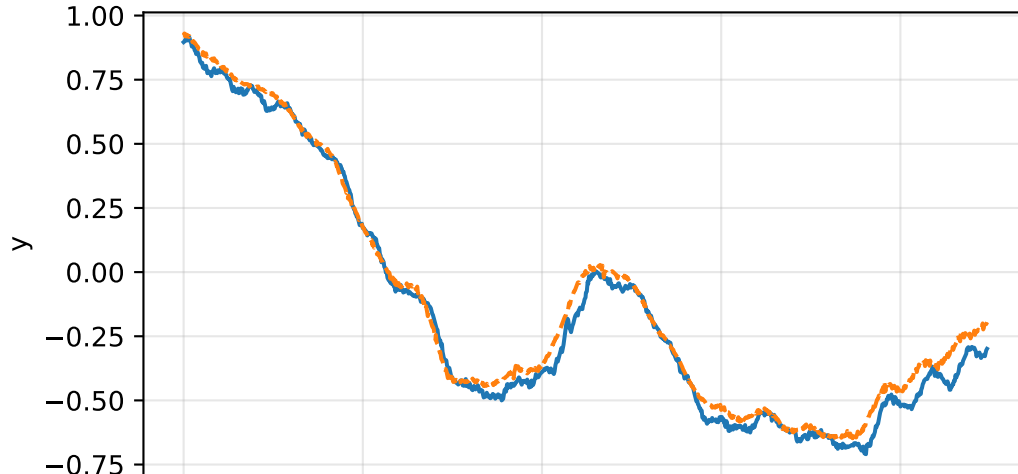
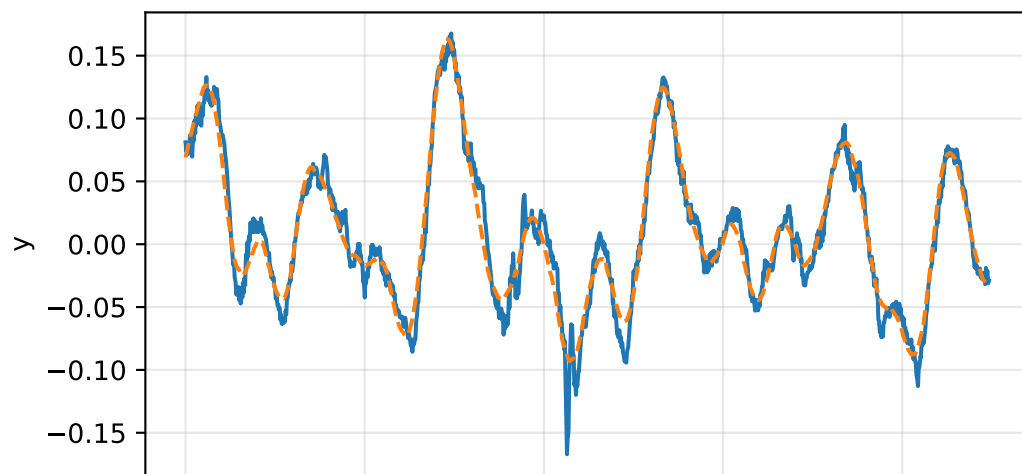
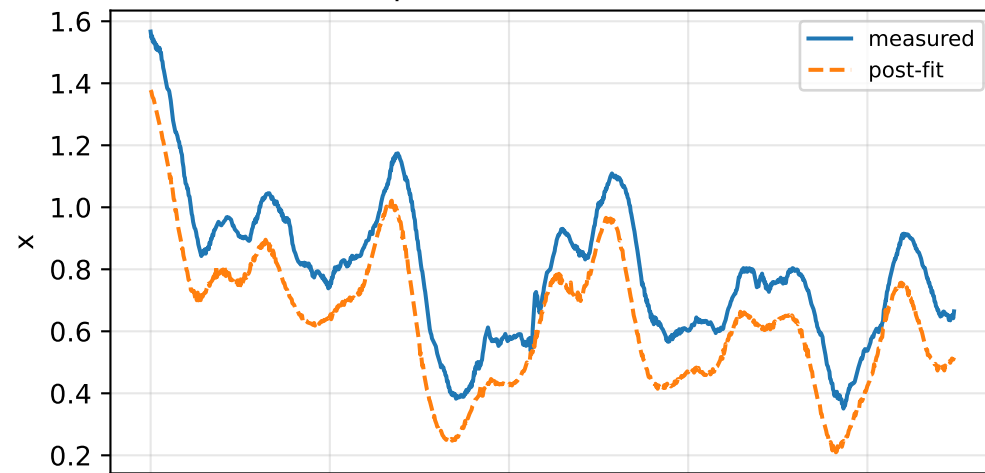


force_over_mass_all_nominal: IMU comparison (diagnostic only)

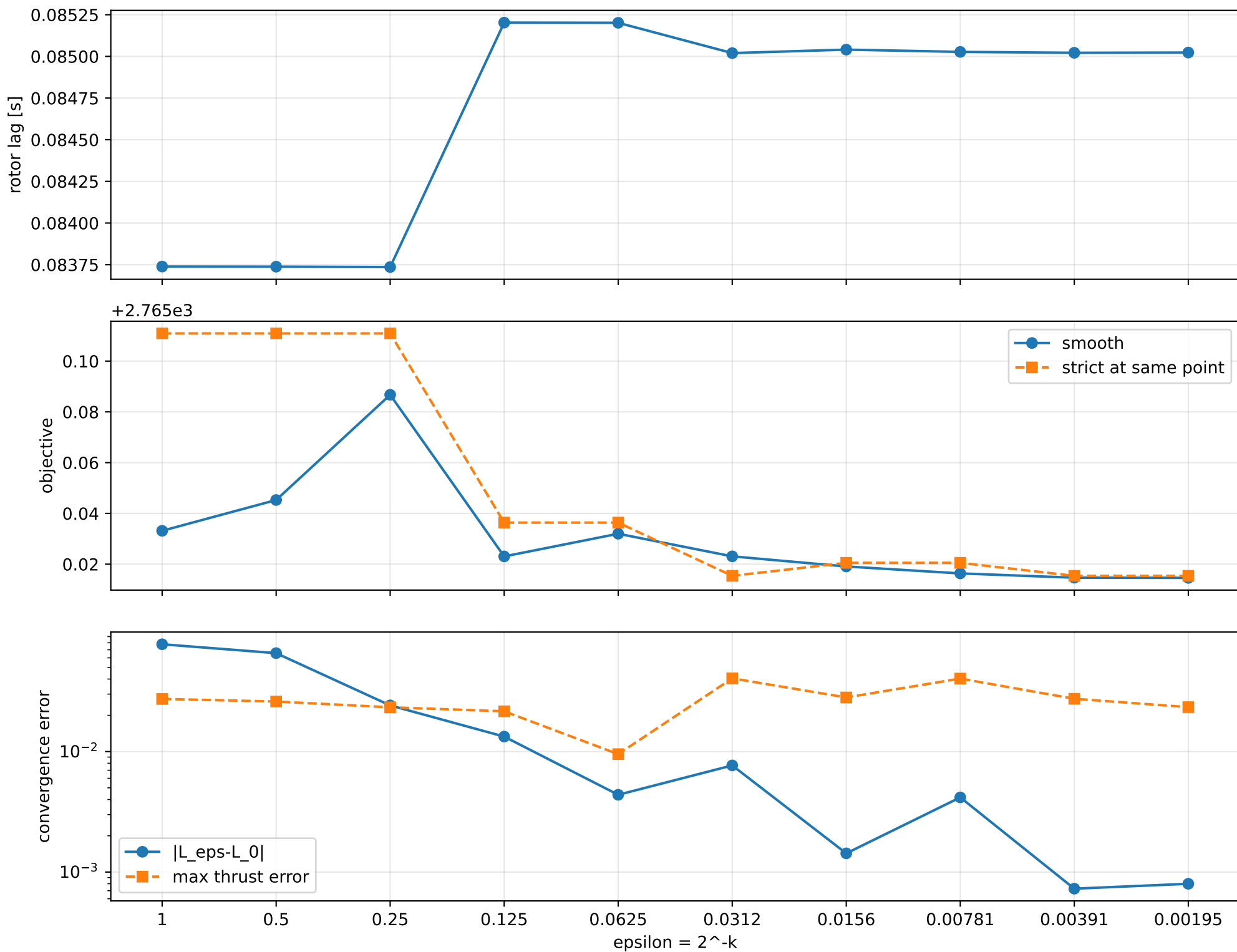
gyro [rad/s]



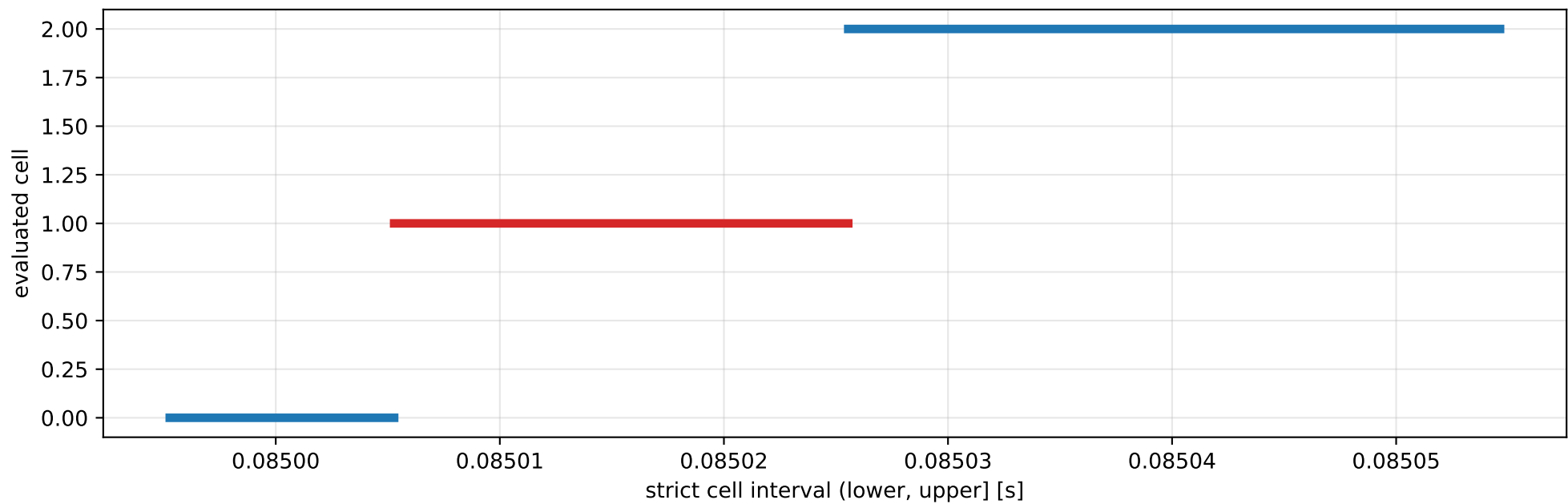
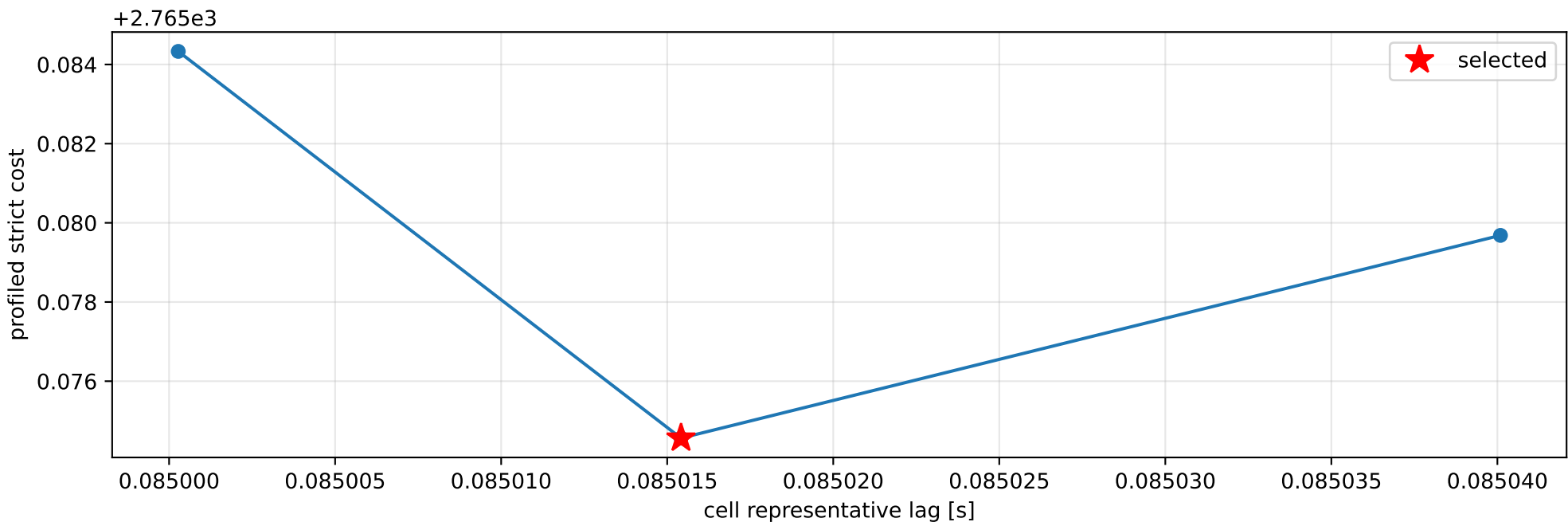
specific force [m/s^2]



force_over_mass_all_nominal: smooth-to-strict continuation



force_over_mass_all_nominal: exact strict-ZOH lag cells



selected (0.0850052834, 0.0850255489] s; final smooth 0.0850230305 s; data boundary=False

force_over_mass_all_nominal: parameter estimate

Case: force_over_mass_all_nominal

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 0.78532251

inertia [kg m²]:

[[2.17408444 -0.24963194 -0.09374229]

[-0.24963194 0.57890252 -0.84679754]

[-0.09374229 -0.84679754 1.92239709]]

CoG body [m]: [-0.01715829 0.0032404 -0.11381419]

force effectiveness: [0.33395137 0.33395151 0.33395145 0.33395132]

scale-free J/m [m²]:

[[2.76839694 -0.31787188 -0.11936788]

[-0.31787188 0.73715258 -1.07827997]

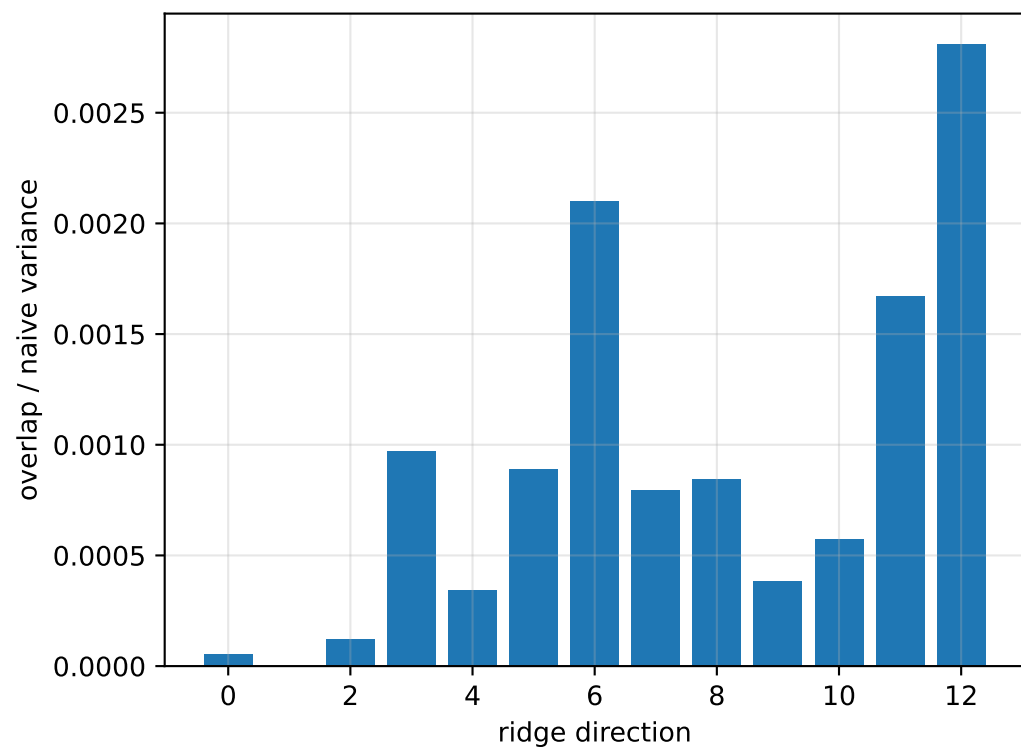
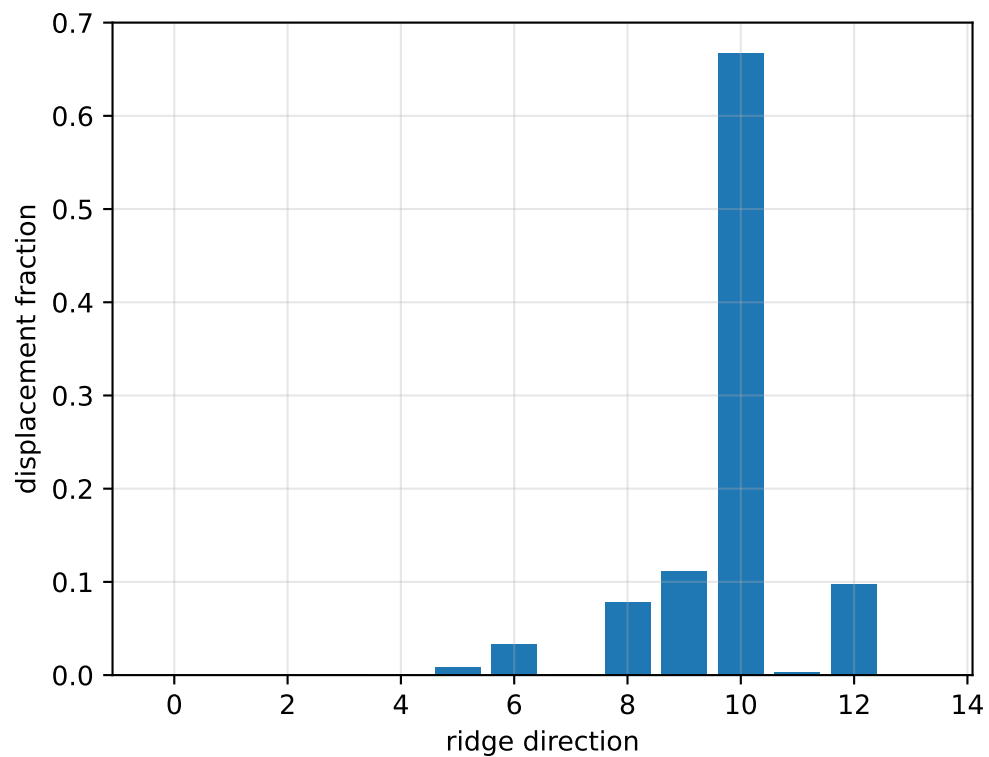
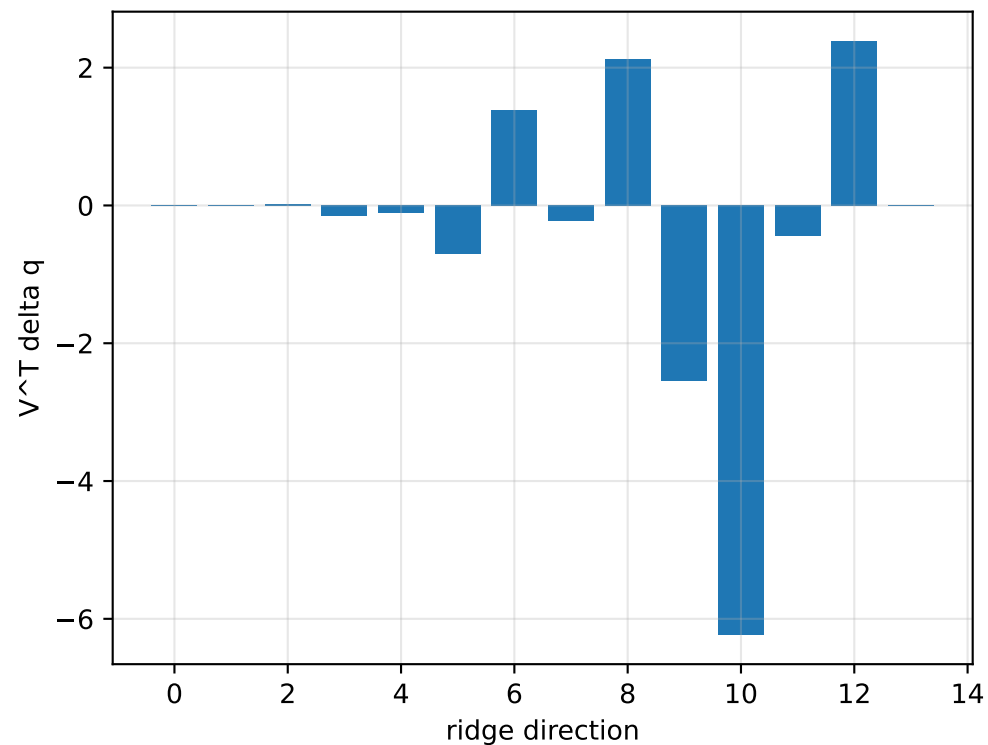
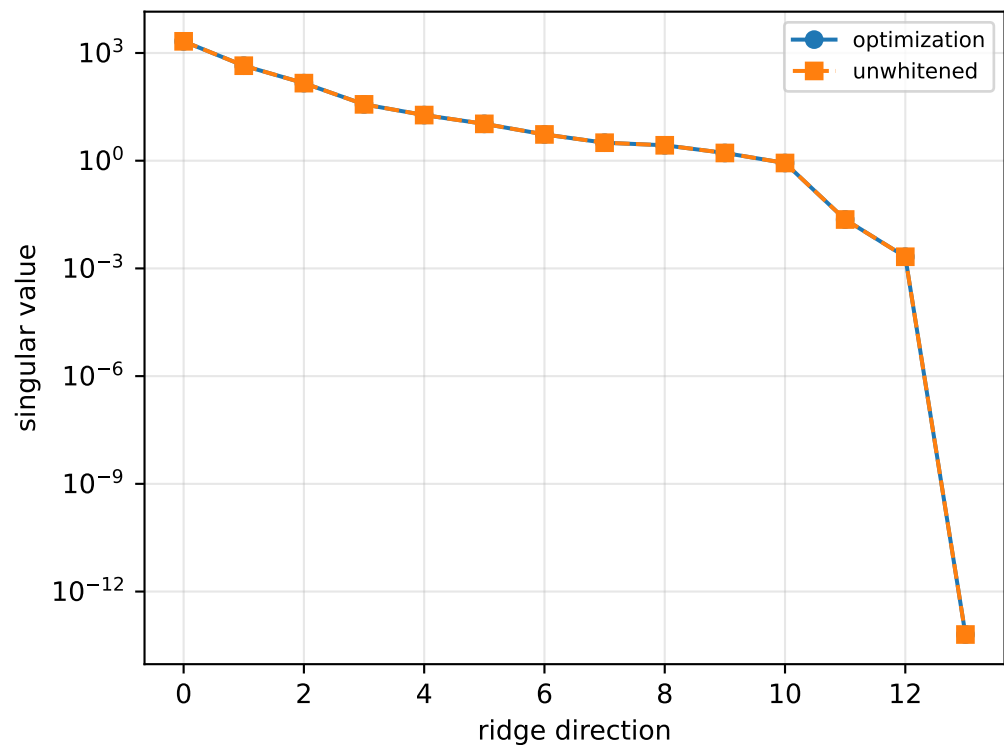
[-0.11936788 -1.07827997 2.44790779]]

scale-free f/m: [0.42524106 0.42524123 0.42524116 0.42524099]

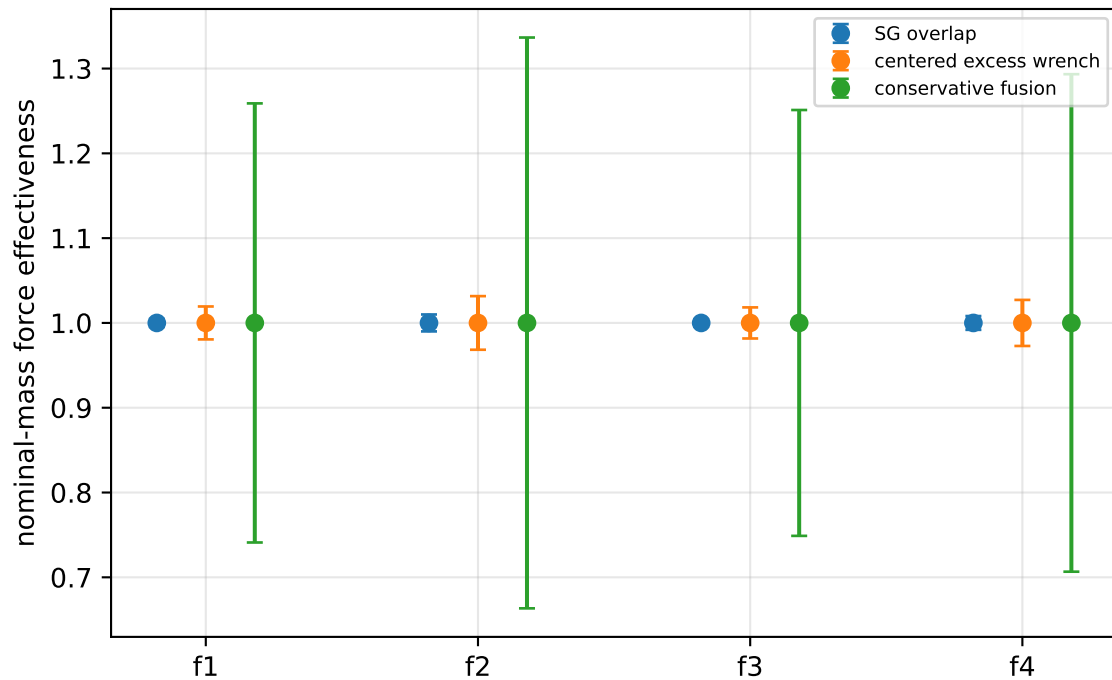
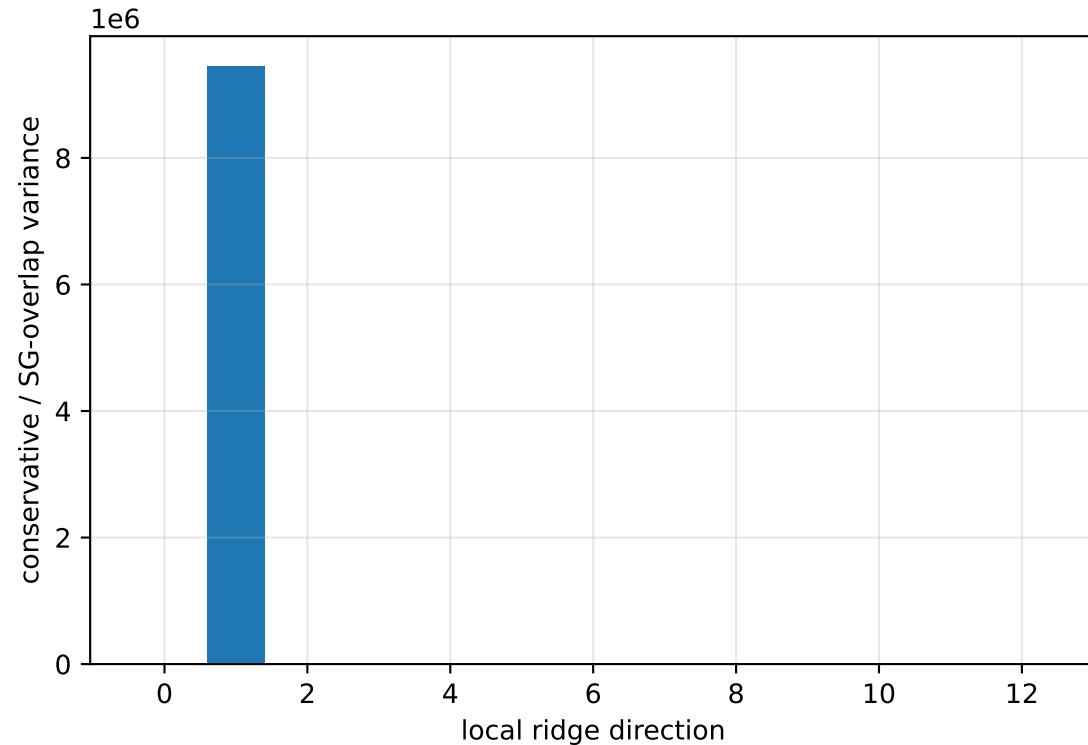
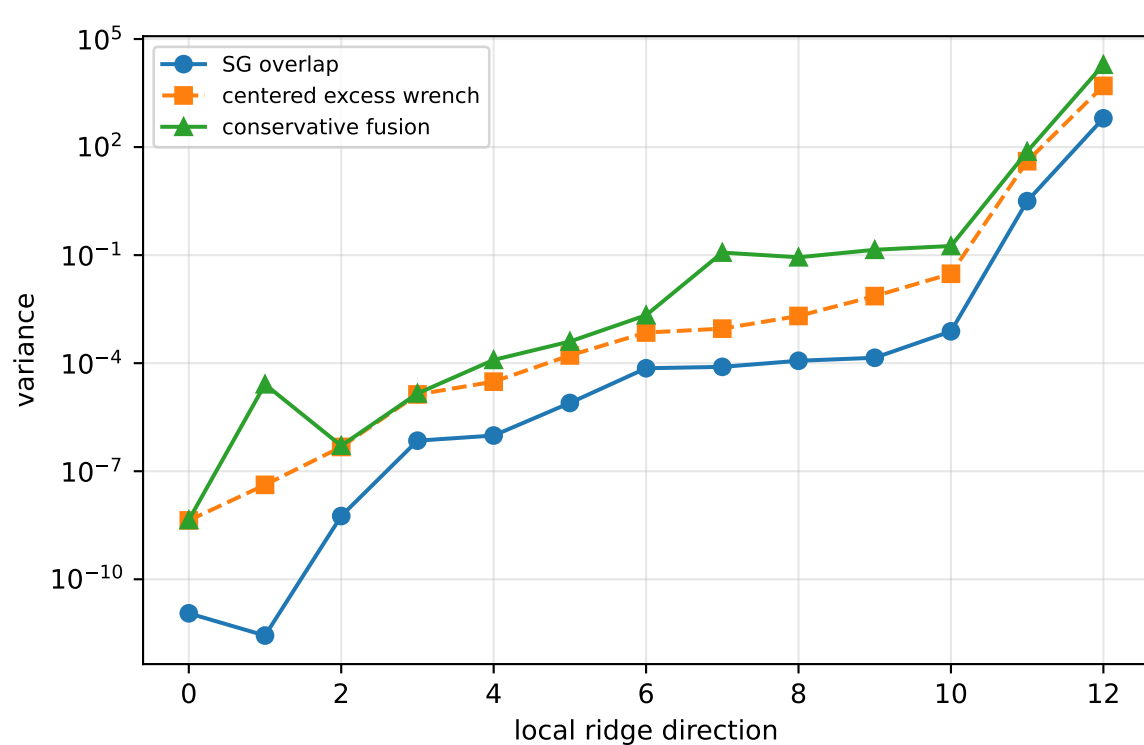
rotor lag cell: (0.0850052834, 0.0850255489) s

representative: 0.0850154161 s

force_over_mass_all_nominal: ridge spectrum (rank 13, nullity 1, ||jv||=7.18e-14)



force_over_mass_all_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 4685$
wrench-to-acceleration recovery max error = $2.66\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

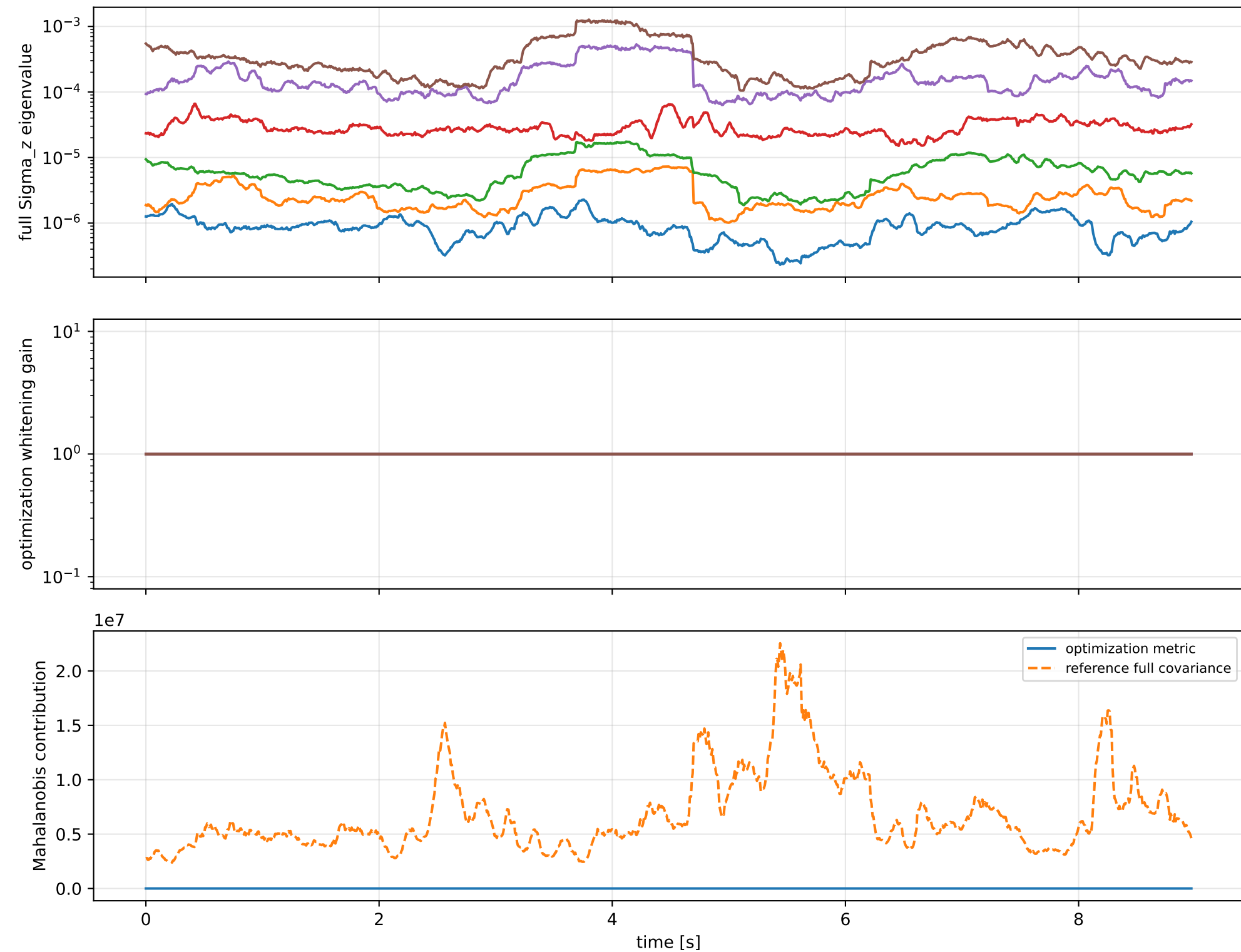
index 12: variance= $1.933\text{e}+04$, ratio=30.72
 $\log_{\text{mass}}=+0.133$; second_moment_diag_1= $+0.0269$
second_moment_diag_2= -0.794 ; second_moment_diag_3= $+0.102$
second_moment_offdiag_12= $+0.46$; second_moment_offdiag_13= -0.0793
second_moment_offdiag_23= $+0.229$; cog_x= $-1.21\text{e-}05$
cog_y= $+2.18\text{e-}05$; cog_z= $-9.4\text{e-}06$
 $\log_{\text{force_eff_1}}=+0.133$; $\log_{\text{force_eff_2}}=+0.133$
 $\log_{\text{force_eff_3}}=+0.133$; $\log_{\text{force_eff_4}}=+0.133$

index 11: variance=74.85, ratio=23.72
 $\log_{\text{mass}}=+0.0126$; second_moment_diag_1= $+0.375$
second_moment_diag_2= -0.416 ; second_moment_diag_3= -0.0223
second_moment_offdiag_12= -0.801 ; second_moment_offdiag_13= $+0.152$
second_moment_offdiag_23= $+0.146$; cog_x= $+6.36\text{e-}05$
cog_y= $-6.54\text{e-}06$; cog_z= -0.000139
 $\log_{\text{force_eff_1}}=+0.0121$; $\log_{\text{force_eff_2}}=+0.0129$
 $\log_{\text{force_eff_3}}=+0.0131$; $\log_{\text{force_eff_4}}=+0.0124$

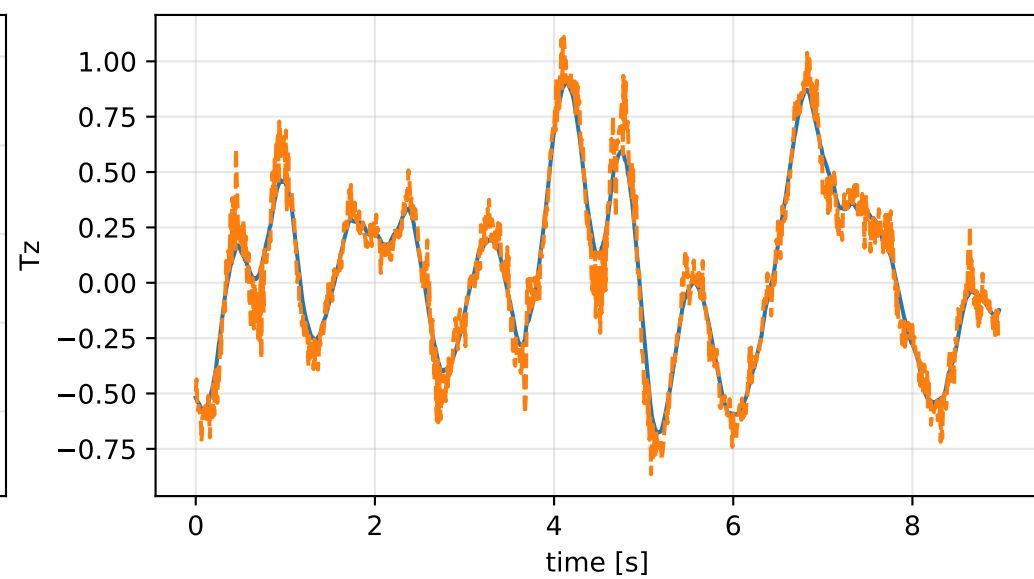
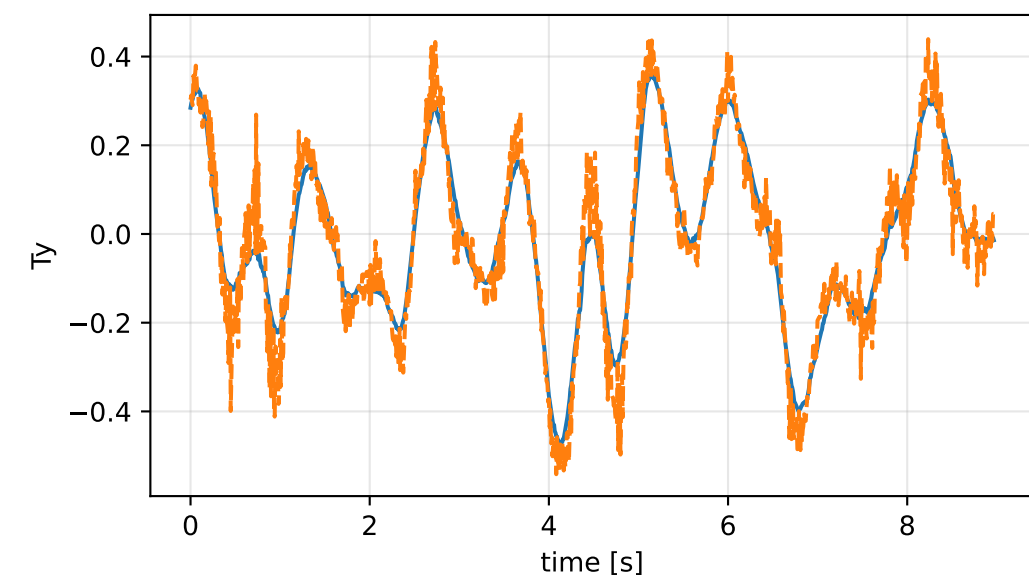
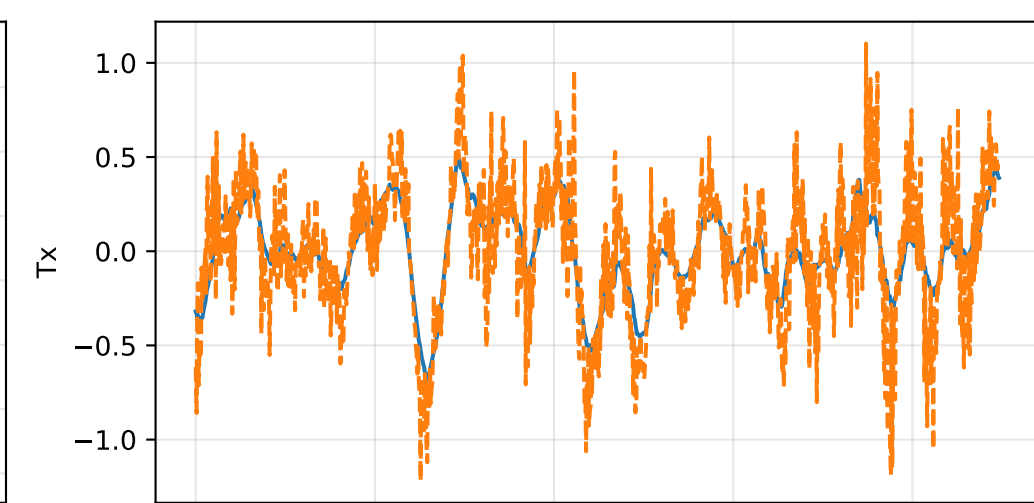
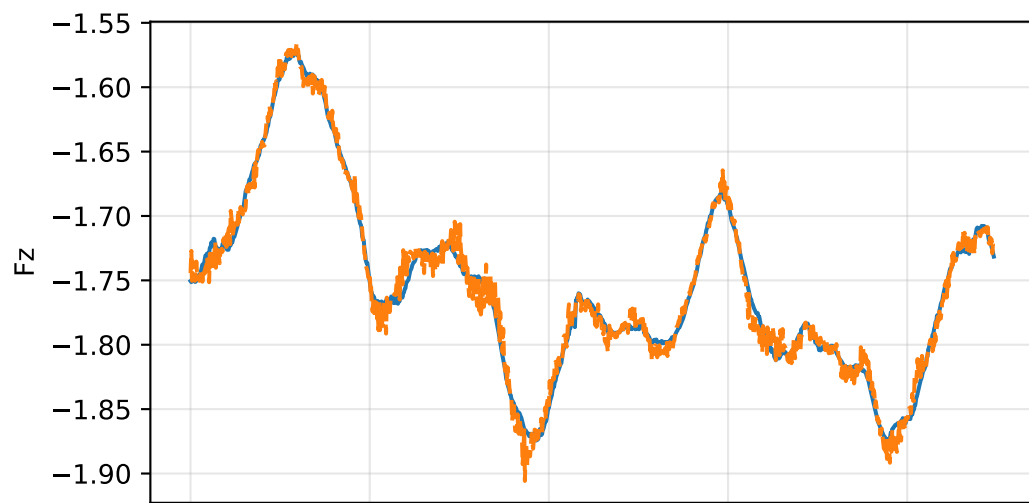
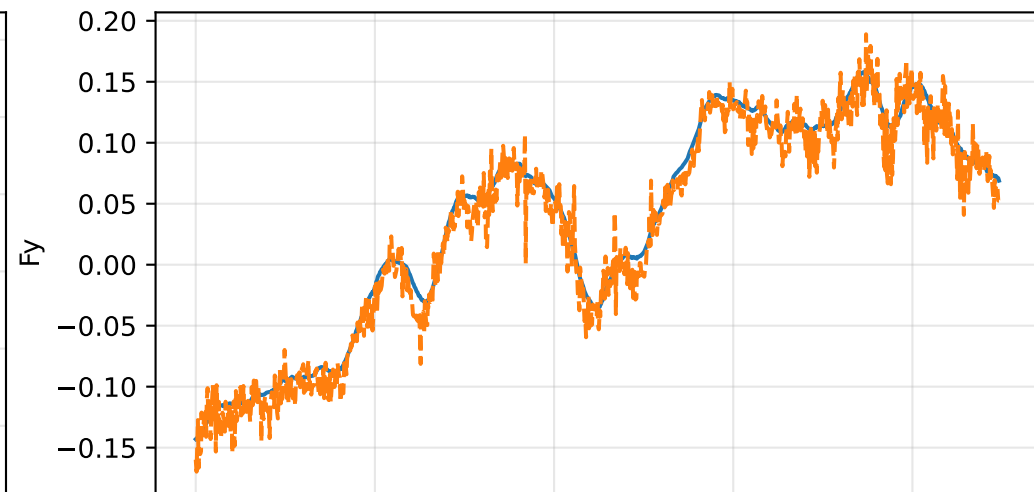
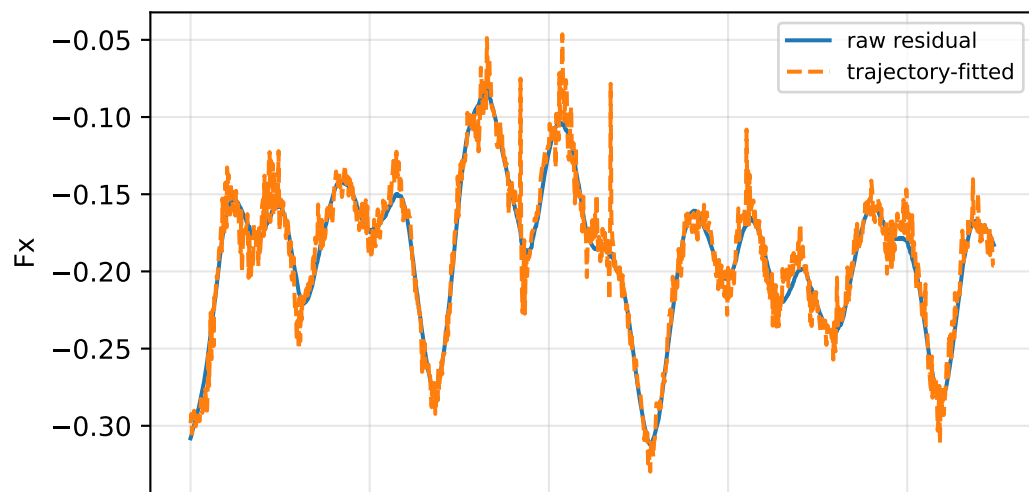
index 10: variance=0.1792, ratio=234
 $\log_{\text{mass}}=+0.131$; second_moment_diag_1= $+0.139$
second_moment_diag_2= $+0.103$; second_moment_diag_3= -0.905
second_moment_offdiag_12= $+0.0685$; second_moment_offdiag_13= -0.0745
second_moment_offdiag_23= $+0.194$; cog_x= $+0.00453$
cog_y= -0.0111 ; cog_z= $+0.0169$
 $\log_{\text{force_eff_1}}=+0.171$; $\log_{\text{force_eff_2}}=+0.209$
 $\log_{\text{force_eff_3}}=+0.0926$; $\log_{\text{force_eff_4}}=+0.0594$

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

force_over_mass_all_nominal: optimization vs reference covariance



force_over_mass_all_nominal: wrench / closure (force max 2.22e-15, torque max 2.22e-16)



force_over_mass_all_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_force_over_mass_all_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/group/force_over_mass_

SHA256: 1299af3479d15bf1e5d1f52aba937e63fa8162cc154062ba161787f233f29dd2

data objective: 2765.01540118

prior objective: 0.0591609049793

total objective: 2765.07456208

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: force_over_mass_all_nominal (force_effectiveness_over_mass)

components: rotor_1, rotor_2, rotor_3, rotor_4

target: [0.42524282 0.42524282 0.42524282 0.42524282]

std: [1.e-05 1.e-05 1.e-05 1.e-05]

final: [0.42524106 0.42524123 0.42524116 0.42524099]

error: [-1.76747460e-06 -1.59668522e-06 -1.66818148e-06 -1.83738438e-06]

standardized residual: [-0.17674746 -0.15966852 -0.16681815 -0.18373844]

factor objective: 0.0591609049793